



VIT RAILS

Project-Abhinav Singh(25BAI10303)

VIT Rails Train Booking System

AIM :-

Our aim through this project given by VITyathri team is to join each campus of our esteemed institute VIT by train network which would ease the student, faculty and other members to travel to each and every branch of VIT over India.

For the same cause we have designed a ticket booking interface for our VIT family members called VITrails.

Overview:-

- *VIT Rails is a console-based train ticket booking system implemented in Python.*
- *It allows users to search for trains between cities.*
- *Book tickets with berth preferences, calculate fares including GST.*
- *Process payments via a simulated card input. The system outputs a formatted ticket upon successful booking.*

Features:-

- *Select source and destination city (from BHOPAL, AMARAVATI, VELLORE, CHENNAI)*
- *Input and validate travel date (must be today or future).*
- *Search available trains on the selected route.*
- *Choose preferred berth type (Lower, Middle, Upper, Side Lower).*
- *Automatic fare calculation with GST.*
- *Simulated payment processing with card number and PIN inputs.*
- *Display booking confirmation with ticket details.*

Data

- *TRAINS: List of available trains with train number, name, source and destination*
- *BASE_FARES: Base fare dictionary for city-to-city routes used in fare calculation*

Usage:-

- 1) *Run the program: python train_booking_system.py.*
- 2) *Enter source city, destination city, and travel date.*

3) Select a train from the search results.

4) Enter preferred berth type.

5) Enter payment card details for simulated payment.

6) Receive booking confirmation with ticket details

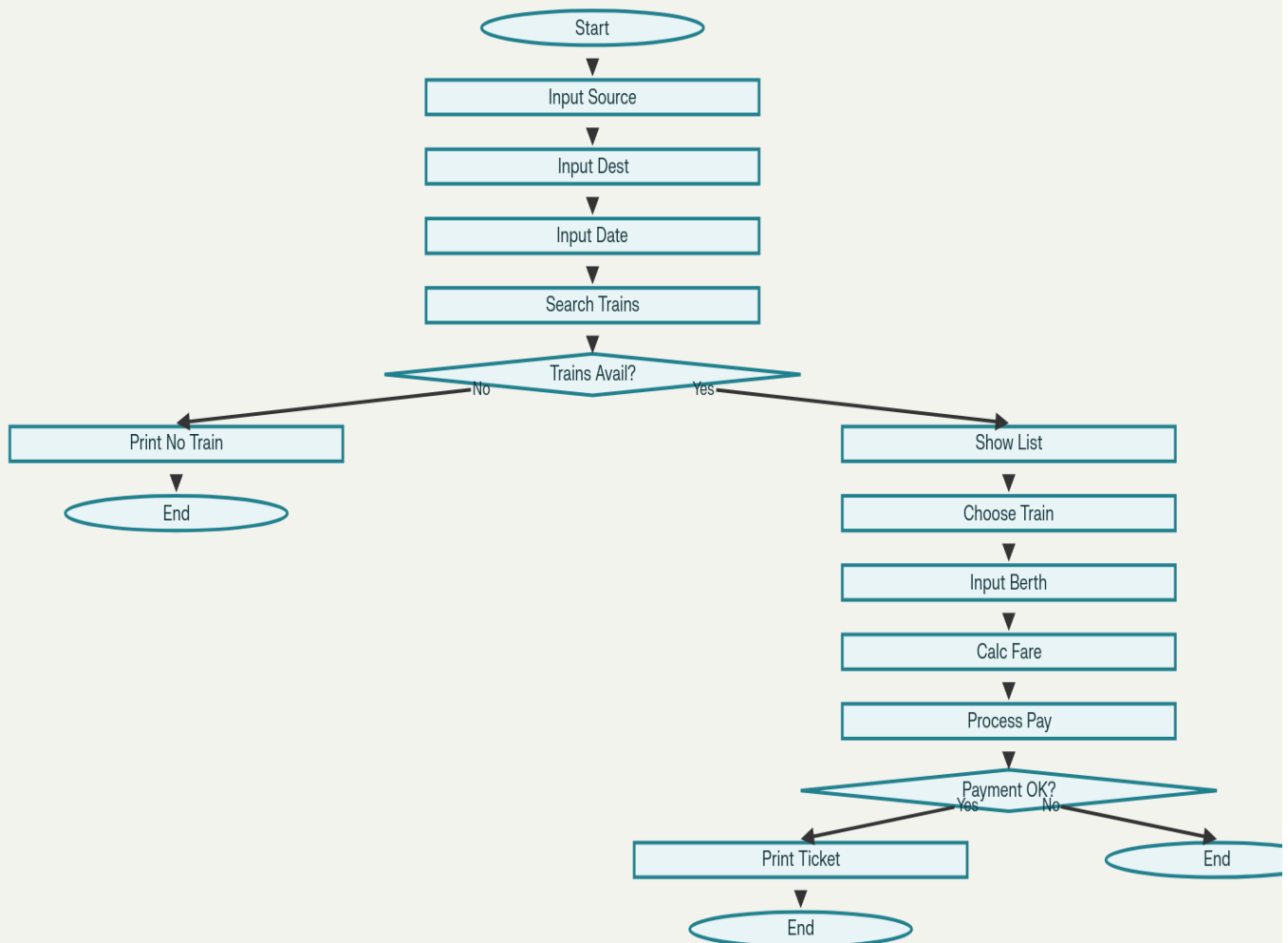
Code Structure.

Code Structure

- *Input validation functions for city and date*
- *Fare calculation including GST*
- *Randomized berth allocation logic*
- *Payment process simulation and validation*
- *Main booking workflow coordinating all steps*

FLOWCHART

Train Booking System Flow



In the flow chart, we have used pseudo code and algorithm.

Algorithm for Train Booking System

1. Start
2. Input Source City
 - Prompt user to enter source city
 - Validate input is not empty and convert to uppercase
3. Input Destination City
 - Prompt user to enter destination city
 - Validate input similarly
4. Input Travel Date
 - Prompt user to enter date in DDMMYYYY format
 - Validate format and check date is today or future
5. Search for Trains on Route
 - Filter available trains with matching source and destination
6. If no trains found
 - Display message "No trains available"
 - End program
7. If trains found
 - Display list of trains with index numbers
 - Prompt user to select train by entering list number
 - Validate selected number is valid
8. Input Preferred Berth Type
 - Show berth options (LB, MB, UB, SL)
 - Validate input is in allowed options
9. Calculate Fare
 - Lookup base fare for source-destination pair

- Compute GST (5% of base fare)
- Calculate total fare = base fare + GST

10. Randomly Allot Coach and Seat Number

- Randomly select coach number (e.g., B1 to B5)
- Randomly select seat number between 1 to 72
- Determine berth code from seat number modulo 8
- logics

11. Process Payment

- Display total amount due
- Prompt for 16-digit card number and validate
- Prompt for 4-digit PIN and validate
- Simulate payment processing delay
- Confirm payment success

12. If payment successful

- Display ticket with train details, travel date, coach, seat no., berth, and fare paid

13. End



This Photo by Unknown Author is licensed under CC BY-SA-NC

CODES

```
def valid_city(x):
    while True:
        c=input(x).strip().upper()
        if c:
            return c
        print("Input Cannot be empty. Please try again")

def valid_date():
    while True:
        try:
            d=input("Enter date of travel(DDMMYYYY): ")
            day=int(d[0:2])
            month=int(d[2:4])
            year=int(d[4:8])
            travel=dt(year, month, day)
            if travel.date()>=dt.today().date():
                return travel
            else:
                print("Travel date must be today or in future.")
        except ValueError:
            print("Invalid date format(Use DDMMYYYY). Please Try Again")
```

- Here we have defined a code for city.
And
- We have defined a code for valid date.


```

def calculate_fare(source, dest, preferred_berth_code=None):
    route_key=(source, dest)
    base_price=BASE_FARES.get(route_key, 0)

    if base_price==0:
        return 0

    GST=base_price*(0.05)
    final_fare=base_price+GST
    return round(final_fare)

def process_payment(fare):
    print("\n-----Payment Gateway-----")
    print(f"Total Amount Due: INR {fare}")

    while True:
        card=input("Enter 16-digit card number:")
        if len(card) == 16 and card.isdigit():
            break
        print("Invalid card number.")

    while True:
        pin= input("Enter 4-digit PIN:")
        if len(pin) == 4 and pin.isdigit():
            break
        print("Invalid PIN:")

    print("\nProcessing payment...")
    time.sleep(3)
    print("Payment successful!")
    return True

```

Here we have defined fare.

And

We have defined process of payment.

```

def booking_system():
    print("-----Welcome to VITrails:Train Booking System-----")

    source=valid_city("Enter Source City : [Ex- BHOPAL, AMARAVATI, VELLORE, CHENNAI]: ")
    dest=valid_city("Enter Destination City : [Ex- BHOPAL, AMARAVATI, VELLORE, CHENNAI]: ")
    travel=valid_date()

    print(f"\n--- Searching for trains from {source} to {dest} ---")
    available_trains=[
        train for train in TRAINS
        if train[2] == source and train[3] == dest]

    if not available_trains:
        print("No trains found for this route. Booking cancelled.")
        print("Available cities in VITrails : BHOPAL, AMARAVATI, VELLORE, CHENNAI.")
        return

    print("Found the following trains:")
    for i, train in enumerate(available_trains):
        print(f"{i+1}. Train No: {train[0]} | Name: {train[1]}")

    while True:
        try:
            choice=int(input("Enter the number of the train you want to book: "))
            if 1<=choice<=len(available_trains):
                selected_train=available_trains[choice - 1]
                break
            else:
                print("Invalid choice. Please enter a number from the list.")
        except ValueError:
            print("Invalid input. Please enter a number.")

```

- Here we have defined booking system.



```

BERTH_TYPES={'LB': 'Lower Berth', 'MB': 'Middle Berth', 'UB': 'Upper Berth', 'SL': 'Side Lower', 'SU': 'Side Upper'}
print("\n-----Seat Preferences-----")
print("Available Preferences: LB , MB , UB , SL ")

while True:
    pref=input("Enter your preferred seat type (LB/MB/UB/SL): ").strip().upper()
    if pref in BERTH_TYPES:
        break
    print("Invalid preference. Please enter LB, MB, UB, SL")

fare=calculate_fare(source, dest)
if fare==0:
    print("\n*ERROR: Could not calculate fare for this route. Booking cancelled.*")
    return

coach=random.choice(['B1','B2','B3','B4','B5'])
seat_no=random.randint(1, 72)
remainder=seat_no%8
if remainder in [1, 4]:
    allotted_berth_code='LB'
elif remainder in [2, 5]:
    allotted_berth_code='MB'
elif remainder in [3, 6]:
    allotted_berth_code='UB'
elif remainder == 7:
    allotted_berth_code='SL'
else:
    allotted_berth_code='SU'
allotted_berth_name=BERTH_TYPES.get(allotted_berth_code, "Unknown Berth")

```

- Here we used else if function and loops in code.

```

if process_payment(fare):
    print("\n --- VITrails Ticket --- ")
    print("_____")
    print(f"Train No: {selected_train[0]} | Name: {selected_train[1]}")
    print(f"Route: {source} --> {dest}")
    print(f"Travel Date: {travel.strftime('%d-%b-%Y')}")
    print("_____")
    print(f"Coach No.: {coach} | Seat No. {seat_no}")
    print(f"Alloted berth: {allotted_berth_name} ({allotted_berth_code})")
    print(f"Pref Type: {BERTH_TYPES.get(pref)}")
    print(f"Fee Paid including GST: INR {fare}")
    print("_____")
    print("Wish you a pleasant journey and\n Travel again with VITrails! ")

if __name__=="__main__":
    booking_system()

```

- After these tickets gets booked and gets print.



This Photo by Unknown Author is licensed under CC BY

Input- Output

```
-----Welcome to VITrails:Train Booking System-----
Enter Source City : [Ex- BHOPAL, AMARAVATI, VELLORE, CHENNAI]: chennai
Enter Destination City : [Ex- BHOPAL, AMARAVATI, VELLORE, CHENNAI]: bhopal
Enter date of travel(DDMMYYYY): 23062025
Travel date must be today or in future.
Enter date of travel(DDMMYYYY): 12122025

--- Searching for trains from CHENNAI to BHOPAL ---
Found the following trains:
1. Train No: 12839 | Name: Raj Bhoj Exp
Enter the number of the train you want to book: 1

-----Seat Preferences-----
Available Preferences: LB , MB , UB , SL
Enter your preferred seat type (LB/MB/UB/SL): 'mb
Invalid preference. Please enter LB, MB, UB, SL
Enter your preferred seat type (LB/MB/UB/SL): ub

-----Payment Gateway-----
Total Amount Due: INR 2152
Enter 16-digit card number:123456789876543
Invalid card number.
Enter 16-digit card number:1234567898765432
Enter 4-digit PIN:1234

Processing payment...
Payment successful!

--- VITrails Ticket ---

-----
Train No: 12839 | Name: Raj Bhoj Exp
Route: CHENNAI --> BHOPAL
Travel Date: 12-Dec-2025

-----
Coach No.: B3 | Seat No. 30
Alloted berth: Upper Berth (UB)
Pref Type: Upper Berth
Fee Paid including GST: INR 2152

-----
Wish you a pleasant journey and
Travel again with VITrails!
```

Hence your ticket is printed with fare.

THANK YOU!!!

By-Abhinav Singh(25BAI10303)